Name: _____

(20 Points)

Part 1:Choose the correct answer

1) The volume of the solid, which is generated when the region bounded by, $y=\cosh x$, $y=\sinh x$, $x=0$ and $x=2$, is rotated a complete revolution about the x–axis, is equal to

A. 2π cubic units
B. 3π cubic units
C. π cubic units
D. 4π cubic units

2. The arc length of the curve of $y=\cosh x$ from $x=0$ to $x=\ln 2$ is equal to

A. $\frac{2}{3}$ length units
B. $\frac{4}{3}$ length units
C. $\frac{3}{2}$ length units
D. $\frac{3}{4}$ length units

3. $\lim_{x\to 0}\left(e^{x}+5x\right)^{\frac{1}{x}}=$

A. e^{4}
B. e^{5}
C. e^{6}
D. e^{7}

4. $\int \cos 5x \sin 3x\, dx =$

A. $\frac{1}{4}\cos 2x-\frac{1}{16}\cos 8x+c$
B. $\frac{1}{4}\cos 2x+\frac{1}{16}\cos 8x+c$
C. $-\frac{1}{4}\cos 2x-\frac{1}{16}\cos 8x+c$
D. $-\frac{1}{4}\cos 2x+\frac{1}{16}\cos 8x+c$

5. Given that $\coth x=4$, then $x=$

A. $\ln\left(\sqrt{\frac{3}{5}}\right)$
B. $\ln\left(\sqrt{\frac{5}{3}}\right)$
C. $\ln\left(\sqrt{\frac{5}{7}}\right)$
D. $\ln\left(\sqrt{\frac{7}{5}}\right)$

6. If $f(x)=\sqrt{\cosh x+\sinh x}$ then $f'(\ln 4)=$

A. 6
B. 4
C. 2
D. 1

7. $\int_{0}^{\pi} x\cos x\, dx =$

A. 2
B. -2
C. π
D. $-\pi$

8. $\int (2x+1)\ln x\ dx =$

A. $(x^2+x)\ln x - \frac{1}{2}x^2 + x + c$

B. $(x^2+x)\ln x + \frac{1}{2}x^2 + x + c$

C. $(x^2+x)\ln x - \frac{1}{2}x^2 - x + c$

D. $(x^2+x)\ln x + \frac{1}{2}x^2 - x + c$

9. $\int \cot^{-1} 4x\ dx =$

A. $x\cot^{-1}4x + \frac{1}{4}\ln\left(1+16x^2\right) + c$

B. $x\cot^{-1}4x + \frac{1}{8}\ln\left(1+16x^2\right) + c$

C. $x\cot^{-1}2x + \frac{1}{4}\ln\left(1+4x^2\right) + c$

D. $x\cot^{-1}2x + \frac{1}{8}\ln\left(1+4x^2\right) + c$

10. The integral $\int_1^\infty \frac{1}{x^p}\,dx$ where p is a constant such that $0 < p < 1$

A. converges to 1

B. converges to $\frac{1}{1-p}$

C. converges to $\frac{1}{p-1}$

D. diverges

11. $\int_0^{\frac{\pi}{4}} \sec^4 x\ dx =$

A. $\sqrt{3}$

B. $2\sqrt{3}$

C. $\frac{4}{3}$

D. $\frac{3}{4}$

12. $\int_0^1 \sqrt{4-x^2}\ dx =$

A. $\frac{\pi}{3} + \frac{\sqrt{3}}{2}$

B. $\frac{2\pi}{3} + \frac{\sqrt{3}}{2}$

C. $\frac{\pi}{3} + \frac{\sqrt{3}}{3}$

D. $\frac{\pi}{6} + \frac{\sqrt{3}}{3}$

Write the suitable partial fraction decomposition (don't compute the values of the constants) of

$$f(x) = \frac{x}{(x-1)(x^3-1)^2}$$

3.

$$f(x) = \frac{x}{(x-1)^3(x^2+x+1)^2}$$

$$\int \frac{x}{(x-1)^3(x^2+x+1)^2} = \int \frac{A}{(x-1)}dx + \int \frac{B}{(x-1)^2}dx + \int \frac{C}{(x-1)^3}dx + \int \frac{Dx+F}{x^2+x+1}dx + \int \frac{Jx+r}{(x^2+x+1)^2}dx$$

